A separable-domain answer for the Lipschitz Malliavin chain rule

Historical partial result for arXiv:1204.2946

Status

This is a reconstructed predecessor packet preserved under the history folder of the full basis-free solution. It is superseded by the parent packet, which removes the separability restriction from the operator-factor conclusion.

Source Question

Pronk and Veraar prove in [1, Proposition 3.10] that if E_0 has a Schauder basis, E_1 is UMD, $p \in (1, \infty)$, and $\phi : E_0 \to E_1$ is Lipschitz, then

$$F \in \mathbb{D}^{1,p}(E_0) \quad \Longrightarrow \quad \phi(F) \in \mathbb{D}^{1,p}(E_1),$$

and there is a bounded linear operator

$$T_F : \gamma(H, E_0) \to L^\infty(\Omega; \gamma(H, E_1))$$

such that $D(\phi(F)) = T_F(DF)$. In Remark 3.11 they write that they do not know whether the assumption that E_0 has a basis can be avoided.

By a density and Hahn-Banach argument this yields $TDF = D\phi(F)$. Hence we can take $T = T_F$. □

Remark 3.11. The first part of the proof is based on the idea in [11, Proposition 5.2], where the result has been proved for Hilbert spaces E_0 and E_1. It is surprising that their argument can be extended to a Banach space setting. We do not know if the assumption that E_0 has a basis can be avoided. In the final part of the argument in [11, Proposition 5.2] a compactness argument is used to construct an operator $T \in L^\infty(\Omega; \mathscr{L}(E_0, E_1))$ such that $D(\phi(F)) = T(DF)$. There seems to be a difficulty in this proof, and at the moment it remains unclear whether such a T exists. Note that there are subtle (measurability) differences between the latter space and $\mathscr{L}(E_0, L^\infty(\Omega; E_1))$ if E_0 and E_1 are infinite dimensional.

4. The divergence operator and the Skorohod integral

Definition 4.1. *Let $p \in [1, \infty)$. Let $\mathrm{Dom}_{p,E}(\delta)$ be the set of $\zeta \in L^p(\Omega; \gamma(H, E))$ for which there exists an $F \in L^p(\Omega; E)$ such that*

$$\mathbb{E} \langle \zeta, DG \rangle_{Tr} = \mathbb{E} \langle F, G \rangle_{E,E^*}, \qquad G \in \mathscr{S} \otimes E^*.$$

In that case, F is uniquely determined, and we write $\delta(\zeta) = F$. The operator δ with domain

Historical Result

Theorem 1. *Let H be a separable Hilbert space, let $p \in (1, \infty)$, let E_0 be an arbitrary Banach space, and let E_1 be a UMD Banach space. If $\phi : E_0 \to E_1$ is Lipschitz with constant L, then for every $F \in \mathbb{D}^{1,p}(E_0)$ one has*

$$\phi(F) \in \mathbb{D}^{1,p}(E_1).$$

Moreover

$$\|D\phi(F)(\omega)\|_{\gamma(H,E_1)} \leq L \|DF(\omega)\|_{\gamma(H,E_0)} \quad \textit{for a.e. } \omega \in \Omega.$$

Theorem 2. *In the setting of Theorem 1, assume in addition that E_0 is separable. Then there is a bounded linear operator*

$$T_F : \gamma(H, E_0) \to L^\infty(\Omega; \gamma(H, E_1))$$

with $\|T_F\| \leq L$ such that

$$D(\phi(F)) = T_F(DF).$$

Corollary 1. *The Schauder-basis assumption in [1, Proposition 3.10] may be replaced by separability of E_0. In particular, no approximation property of E_0 is needed for the full operator-factor conclusion in the separable case.*

Proof Intuition

The source proof uses a basis to approximate the identity on E_0 by uniformly bounded finite-dimensional projections. This is stronger than what the chain rule needs. A Malliavin-Sobolev random variable is already obtained as a limit of smooth random variables with finite-dimensional ranges. We can therefore apply the finite-dimensional Lipschitz chain rule to those approximants and pass to the limit.

The remaining operator T_F is a factorization of the already constructed derivative through DF. Once the pointwise estimate $\|D\phi(F)\| \leq L\|DF\|$ is known, separability of $\gamma(H, E_0)$ gives a measurable norming functional for DF. This norming functional produces a rank-one operator field that maps $DF(\omega)$ to $D\phi(F)(\omega)$ with norm at most L.

A finite-dimensional lemma

Lemma 1. *Let V be a finite-dimensional Banach space, let E_1 be UMD, and let $\psi : V \to E_1$ be Lipschitz with constant L. If $G \in \mathbb{D}^{1,p}(V)$, then $\psi(G) \in \mathbb{D}^{1,p}(E_1)$ and*

$$\|D\psi(G)(\omega)\|_{\gamma(H,E_1)} \leq L\|DG(\omega)\|_{\gamma(H,V)} \quad a.s.$$

Proof. Identify V linearly with $\mathbb{R}^m$ and convolve ψ with a standard smooth mollifier on V. This gives C^∞ maps $\psi_\varepsilon : V \to E_1$ such that

$$\|\psi_\varepsilon(x) - \psi(x)\| \to 0 \quad (x \in V), \qquad \|\psi'_\varepsilon(x)\|_{\mathcal{L}(V,E_1)} \leq L.$$

The smooth chain rule [1, Proposition 3.9] gives

$$\psi_\varepsilon(G) \in \mathbb{D}^{1,p}(E_1), \qquad D\psi_\varepsilon(G) = \psi'_\varepsilon(G)DG,$$

and the ideal property of γ-radonifying operators gives the pointwise bound

$$\|D\psi_\varepsilon(G)(\omega)\|_{\gamma(H,E_1)} \leq L\|DG(\omega)\|_{\gamma(H,V)}.$$

Also $\psi_\varepsilon(G) \to \psi(G)$ in $L^p(\Omega; E_1)$. Since E_1 is UMD, $\gamma(H, E_1)$ is reflexive, so the compactness lemma [1, Lemma 3.8] implies $\psi(G) \in \mathbb{D}^{1,p}(E_1)$, after passing to a weakly convergent derivative subsequence.

Mazur's lemma gives convex combinations of the derivatives $D\psi_{\varepsilon_j}(G)$ which converge strongly in $L^p(\Omega; \gamma(H, E_1))$ to $D\psi(G)$. The same pointwise bound is preserved by convex combinations. Passing to an almost surely convergent subsequence yields the asserted pointwise estimate. □

Proof of Theorem 1

Choose smooth random variables $F_n \in \mathscr{S} \otimes E_0$ such that

$$F_n \to F \quad \text{in } \mathbb{D}^{1,p}(E_0).$$

Each F_n takes values in a finite-dimensional subspace $V_n \subseteq E_0$. Applying Lemma 1 to $\phi|_{V_n} : V_n \to E_1$ gives

$$\phi(F_n) \in \mathbb{D}^{1,p}(E_1), \qquad \|D\phi(F_n)(\omega)\|_{\gamma(H,E_1)} \leq L\|DF_n(\omega)\|_{\gamma(H,E_0)}.$$

Since ϕ is Lipschitz,

$$\phi(F_n) \to \phi(F) \quad \text{in } L^p(\Omega; E_1).$$

The derivatives $D\phi(F_n)$ are bounded in

$$L^p(\Omega; \gamma(H, E_1)),$$

so the compactness lemma again gives $\phi(F) \in \mathbb{D}^{1,p}(E_1)$ and, after passing to a subsequence,

$$D\phi(F_n) \rightharpoonup D\phi(F) \quad \text{in } L^p(\Omega; \gamma(H, E_1)).$$

It remains to keep the pointwise estimate. By Mazur's lemma there are convex combinations

$$Z_k = \sum_{n \geq k} a_{k,n} D\phi(F_n)$$

which converge strongly in $L^p(\Omega; \gamma(H, E_1))$ to $D\phi(F)$. Set

$$M_k = \sum_{n \geq k} a_{k,n} \|DF_n\|_{\gamma(H,E_0)}.$$

Since $DF_n \to DF$ in $L^p(\Omega; \gamma(H, E_0))$, we have

$$M_k \to \|DF\|_{\gamma(H,E_0)} \quad \text{in } L^p(\Omega).$$

For every k,

$$\|Z_k(\omega)\|_{\gamma(H,E_1)} \leq LM_k(\omega) \quad \text{a.s.}$$

Taking an almost surely convergent subsequence of Z_k and M_k gives

$$\|D\phi(F)(\omega)\|_{\gamma(H,E_1)} \leq L\|DF(\omega)\|_{\gamma(H,E_0)}$$

for almost every ω. This proves Theorem 1.

Proof of Theorem 2

Put

$$G = \gamma(H, E_0), \qquad Y = \gamma(H, E_1), \qquad X(\omega) = DF(\omega), \qquad Z(\omega) = D\phi(F)(\omega).$$

If E_0 and H are separable, then G is separable because finite-rank operators with coefficients from countable dense subsets are dense in $\gamma(H, E_0)$.

We use the standard measurable norming-functional selection for separable Banach spaces: there is a Borel map

$$J : G \to B_{G^*}$$

such that $Jx(x) = \|x\|$ for all $x \in G$. This follows from the Kuratowski–Ryll-Nardzewski selection theorem applied to the compact-valued map $x \mapsto \{x^* \in B_{G^*} : x^*(x) = \|x\|\}$, where B_{G^*} is equipped with the weak-star topology.

For $R \in G$, define

$$(T_F R)(\omega) = \begin{cases} J(X(\omega))(R)\, Z(\omega)/\|X(\omega)\|, & X(\omega) \neq 0, \\ 0, & X(\omega) = 0. \end{cases}$$

The map $\omega \mapsto J(X(\omega))(R)$ is measurable for every fixed R, and Z is strongly measurable, so $T_F R$ is strongly measurable as a Y-valued random variable. The pointwise estimate from Theorem 1 gives

$$\|(T_F R)(\omega)\|_Y \leq L\|R\|_G$$

almost surely; hence T_F maps G boundedly into $L^\infty(\Omega; Y)$ and $\|T_F\| \leq L$. Linearity in R is pointwise linearity of $J(X(\omega))$. Finally,

$$(T_F X)(\omega) = J(X(\omega))(X(\omega))\, Z(\omega)/\|X(\omega)\| = Z(\omega)$$

when $X(\omega) \neq 0$, while if $X(\omega) = 0$ then $Z(\omega) = 0$ by the same pointwise estimate. Therefore $T_F(DF) = D\phi(F)$.

Historical Limitation

This predecessor removed the Schauder-basis and bounded approximation property requirements in the separable version of the source proposition, and proved the membership theorem for arbitrary E_0. It remained partial because the operator-factor conclusion was proved only under separability of E_0. The parent packet supersedes this limitation.

References

[1] M. Pronk and M. Veraar, *Tools for Malliavin calculus in UMD Banach spaces*, Potential Analysis 40 (2014), 307–344; arXiv:1204.2946.